ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q1, Q3, Q4

Question 1

(a) (i) $f_x = f / p = 16\text{ mm} / 5\ \mu\text{m} = 16\times10^{-3} / 5\times10^{-6} =$ **3200 px**.

(ii) Sensor width $W_s = 1280 \times 5\ \mu\text{m} = 6.4$ mm. By similar triangles FOV $= W_s \cdot Z / f = 6.4 \times 400 / 16 =$ **160 mm**.

Distance per pixel = FOV / 1280 = 160/1280 = **0.125 mm/px** (chek: $Z/f_x = 400/3200 = 0.125$ mm/px).

(b) LSB $= V_{range}/2^n = 5\text{ V} / 2^{12} = 5/4096 = 1.2207$ mV.

Smallest resolvable force ΔF = LSB / S = 1.2207 mV / (10 mV/N) = **0.1221 N**.

(c) $F = e^{AT} = e^{-0.2} = 0.8187$. $G = A^{-1}(F - I)B = (0.8187 - 1)/(-2) = 0.0906$.

Hence the exact model is $x_{k+1} =$ **$0.0906\ x_k + 0.8187\ u_k$**.

Question 3

(a) (1) Differentiation is a high-pass operation (gain $\propto \omega$), so encoder measurement/quantisation noice is amplifed, by about $1/T = 1000$ at 1 kHz.

(2) Encoder quantisation: $\Delta\theta$ is a whole number of counts, so $\hat{\omega} = \Delta\theta/T$ moves in steps of (count size)/T; at low speed the estimate is a coarse staircase of zeros and spikes.

(b) $H(s) = \omega_c/(s + \omega_c)$, $\omega_c = 2\pi(20) = 125.66$ rad/s. Tustin: $s = (2/T)(z - 1)/(z + 1)$, T = 1 ms. With $a = \omega_c T/2 = 0.06283$: $H(z) = a(1 + z^{-1}) / [(1 + a) - (1 - a)z^{-1}]$.

$y_k = [a/(1+a)](u_k + u_{k-1}) + [(1-a)/(1+a)]\ y_{k-1} =$ **$0.05912\ (u_k + u_{k-1}) + 0.88177\ y_{k-1}$**.

DC gain: $z = 1$ gives $H(1) = 2(0.05912)/(1 - 0.88177) = 0.11824/0.11823 = 1.000$ ✓.

(c) $A - LC = [-\ell_1, 1; -\ell_2, -2]$; characteristic polynomal $\det(sI - A + LC) = (s + \ell_1)(s + 2) + \ell_2 = s^2 + (2 + \ell_1)s + (2\ell_1 + \ell_2)$.

Desired $(s + 10)^2 = s^2 + 20s + 100$. Then $2 + \ell_1 = 20 \Rightarrow \ell_1 = 18$, and $2\ell_1 + \ell_2 = 100 \Rightarrow \ell_2 = 100 - 18 = 82$.

$L = [18, 82]^T$.

(d) Advantage of the observer: it uses the plant model and the applied input u, so its velocity estimate has (almost) no phase lag; the 20 Hz filter on differenced data always delays by about $1/(2\pi f_c) \approx 8$ ms. It can also reconstruct states that cannot be obtained by differencing (e.g. disturbance, when augmented).

Fliter is safer when the model is uncertain or changes (unknown payload, friction, contact, actuator saturation): the observer then gives a biased or poorly-behaved estimat while the model-free filter still follows the measured data.

Question 4

(a) Substituting u: $\ddot{\theta} + (a + bk_2)\dot{\theta} + bk_1\theta = bN_{ff}\ r$ (the c term is the gravity load, handled separately).

So $s^2 + 4s + 2 = 0$, giving $s = -0.586$ and $s = -3.414$. DC gain $= bN_{ff}/(bk_1) = N_{ff}/2 = 1 \Rightarrow$ **$N_{ff} = 2$**.

(b) With $b = 0.75$ ($N_{ff} = 8$, $k_1 = 2$): $G_{DC} = bN_{ff}/(c + bk_1) = 0.75 \times 8/(6 + 0.75 \times 2) = 6/7.5 = 0.8$.

θ_{ss} = **0.8**. Tracking error = (1 − 0.8)/1 × 100 = **20%**. (Less than the 25% actuator loss, because the stiffness also dropped.)

(c) Substituting u into the plant and using $\dot{\xi} = -\theta$ (r = 0): characteristic polynomial $s^3 + (a + k_2)s^2 + (c + k_1)s + K_I$.

Desiered: $(s + 4)(s^2 + 4s + 8) = s^3 + 8s^2 + 24s + 32$.

$2 + k_2 = 8 \Rightarrow$ **$k_2 = 6$**; $6 + k_1 = 24 \Rightarrow$ **$k_1 = 18$**; **$K_I = 32$**.

(d) In stedy state $\dot{\xi} = r - \theta_{ss} = 0$, so $\theta_{ss} = r$ whatever b and c are, provided the loop is stabel.

Failure mode: integrater windup — when the actuator saturates the integral keeps growing and causes a big overshoot later.